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by clean energy**

	COMPANY NAME: Inentec
	END USE LOCATION: USA
	QUOTATION #: H17-1016 rev 01
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April 28, 2017

Dieter Olson
Inentec, Inc.
509-946-8954

Xebec Quotation Number: H17-1016 rev 01

RE: PROPOSAL FOR XEBEC G2 PSA UNIT

Dear Dieter,

We thank you for your interest in Xebec's PSA technology. For the process conditions listed below, we are pleased to propose one (1) Xebec PSA3H-G2-330A-480 at a price of \$93,000 USD. We are pleased to submit our proposal in the following format:

Section 1: System Components
Section 2: Financial Proposal
Section 3: General Terms and Conditions of Sale

The proven advantages of the Xebec G2 PSA system are:

- **Compact System** – Fast PSA cycle that allows adsorber vessels to be 5-15 times smaller than the conventional PSA.
- **Skid Mounted Unit, Pre-loaded with Adsorbents** – Unlike conventional PSAs, field-erection and adsorbent loading is not required for installation.
- **Highly Reliable Rotary Valves (typically <1 CPM)** – Long seal wear life, with 5-Year intervals between servicing.
- **Advanced PSA Nine Adsorber Bed Design** – Triple equalization PSA cycle, mechanically set to ensure the optimal cycle design.
- **Single Input for Rotary Valve and PSA Cycle Speed** – Simple to operate and requires minimal signals.
- **Optimal Process Cycle Design for Opening & Closing of Ports** – Gradual gas flow changes resulting in very long adsorbent life.

Xebec G2 PSA performance specs shown below (Section 1.1) are estimates based on past experience and testing at similar conditions.

We trust that we have interpreted your requirements correctly.

Yours sincerely,

Gary Blizzard

Gary Blizzard
Sales Manager, USA

SECTION 1: SYSTEM COMPONENTS

1.1 DESIGN BASIS

Stated performance is based on the Purchaser providing the stated nominal feed gas composition with deviations of not more than $\pm 2.5\%$ relative change, and provided the stated maximum exhaust pressure is not exceeded:

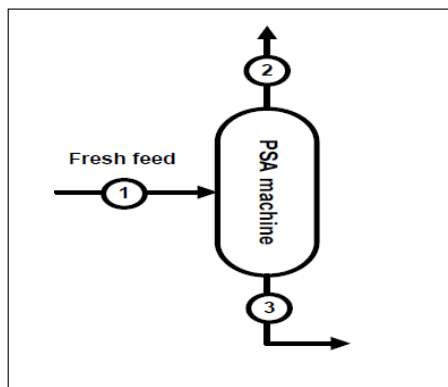


Figure 1. Schematic diagram of Hydrogen removal system.

Flow case 1: High pressure

H2 purity %	99.999
H2 expected recovery %	73

Table 1. Stream properties, flowrates, and compositions.

Stream	1	2	3
	Fresh feed	Product	Exhaust
Flow (NCMH)	35	8	27
Flow (scfm)	22	5	17
P (psig)	290	275	2
Temperature (°C)	40	40	40
%Volume			
Stream	1	2	3
H2	29.7	99.999	10.2
CH4	0.0	Balanced	0.0
CO2	21.2	Balanced	27.1
CO	29.7	Balanced	37.9
N2	19.0	Balanced	24.3
H2O	0.4	Balanced	0.4
Total	100.0	100.0	100.0

PSA GAS PURIFICATION SYSTEM

Flow case 2: Low pressure

H2 purity %	99.999
H2 expected recovery %	70

Table 1. Stream properties, flowrates, and compositions.

Stream	1	2	3
	Fresh feed	Product	Exhaust
Flow (NCMH)	22	5	18
Flow (scfm)	14	3	11
P (psig)	160	145	2
Temperature (°C)	40	40	40
%Volume			
Stream	1	2	3
H2	29.6	99.999	11.2
CH4	0.0	Balanced	0.0
CO2	21.2	Balanced	26.7
CO	29.6	Balanced	37.4
N2	19.0	Balanced	23.9
H2O	0.6	Balanced	0.8
Total	100.0	100.0	100.0

PROCESS CONDITIONS

- Exhaust & Product Temperatures: $\pm 10^{\circ}\text{C}$ from Feed Temperature expected but highly dependent on ambient Temperature.
- Water: Design basis of 100% relative humidity (RH) with zero liquid water in feed gas to the PSA.
- Flow units of NCMH are at 0°C and 101.3 KPa(a). Flow units of SCFM are at 15.6°C and 14.7 PSIA.
- Zero particulate matter over 0.5 micron in size in the feed stream.
- Guaranteed conditions given are at the nominal conditions stated above.

PSA GAS PURIFICATION SYSTEM

1.2 STANDARD FEATURES & SPECIFICATIONS

STANDARD FEATURES & SPECIFICATIONS	FEATURES		
	- Advanced 9-Bed Fast-Cycle PSA Process with Three Equalizations - Patent-Protected, Multi-Port Rotary Control Valves for Robust & Reliable Operation - Compact, Modular, & Skid-mounted Design with Pre-loaded Adsorbents for Quick & Simple Installations in Space-Restricted Layouts - Xebec standard painting and finishes used throughout the PSA as required for industrial environment - Minimal Start-Up Time to Full Load Conditions - Adjustable Cycle Speed that Maximizes Performance with Changing Process Parameters		
	MECHANICAL		
		MINIMUM	MAXIMUM
	Operating pressure:	0 Barg (0 Psig)	22.8 Barg (330 Psig) MAWP
	Operating Ambient Temperature:	4°C (40°F)	40°C (104°F)
	Process Operating Temperature:	4°C (40°F)	60°C (140°F)
	Control Panel Operating Temperature:	0°C (32°F)	40°C (104°F)
	Storage Temperature:	-20°C (-4°F)	48°C (118°F)
	Standard Xebec MOC will be used to suit process conditions Process connections and battery limits will be inside Xebec standard skid boundaries		
	ELECTRICAL		
		Classification	ELECTRICAL POWER SUPPLY
	Variable Speed Drive (VSD) located in Control Panel* (CP1):	Non hazardous area	460/575 VAC 3Ph 60 Hz OR 120 VAC 1 Ph for PSA motor, 24 VDC for controls and instruments
	UL Certified Motor:	Class I div.2 group B or D	Power supply from CP1
	Junction Box - Prox. Switch (JB1):	Class I div.2 group B or D	24 VDC
	Proximity Switch:	Class I div.2 group B or D	Power supply from customer's IS barrier through JB1
	Variable Speed Drive (VSD) located in Control Panel* (CP1):	Non hazardous area	460/575 VAC 3Ph 60 Hz OR 120 VAC 1 Ph for PSA motor, 24 VDC for controls and instruments
	DESIGN CODES & SPECIFICATIONS		
		Protection Rating:	NEMA 4 steel panel, internal components suitable for class 1 div 2 group B (methane) or D (hydrogen)
	North American standards:	CEC C22.1, NEC, NFPA 70 - National Electric Code, NFPA 70, C22.1, NFPA70 (HazLoc-NA) Ex protection: HazLoc-NA, Class I (gas), Division 2 Group B (H2) or D such as AExia/b, AExd, AExe and others as required	
	Control Panel:	NEMA 4 steel panel	
	Power Consumption:	2 Amps (10 Amps max.)	
	Protection Rating:	NEMA 4 steel panel, internal components suitable for class 1 div 2 group B (methane) or D (hydrogen)	
	North American standards:	CEC C22.1, NEC, NFPA 70 - National Electric Code, NFPA 70, C22.1, NFPA70 (HazLoc-NA) Ex protection: HazLoc-NA, Class I (gas), Division 2 Group B (H2) or D such as AExia/b, AExd, AExe and others as required	
STANDARD FEATURES &	CERTIFICATIONS		
	UL Certified Electrical Components:	North American Model Only	

PSA GAS PURIFICATION SYSTEM

SPECIFICATIONS (cont.)	GENERAL ARRANGEMENT	
	Preliminary Plot Plan:	1.8m x 1.9m x 2.8m high (5.8' x 6.3' x 9.3' high)
	Estimated weight:	1300 Kg (2900 lbs)
	Required Foundation:	Concrete pad to be provided by Purchaser

Note: All dimensions, weights and sizes shown here are for information purposes only and may vary depending on model capacity and options. Standard general arrangement, instruments, and components are provided for in this proposal.

1.3 INCLUDED EQUIPMENT & AVAILABLE OPTIONS

Selected options in this proposal are given in Section 2: Financial Proposal. Detail descriptions of the included equipment and available options are provided in this section.

1.2.1 Included Equipment:

INCLUDED EQUIPMENT	AUTOMATIC (STANDARD)	
	This package includes all equipment necessary for safe, unattended, auto-control and turndown operation of the PSA when connected to the Purchaser's PLC with Purchaser's PLC programming approved by Xebec.	

A detail description of the included equipment is given below:

PSA GAS PURIFICATION SYSTEM

INCLUDED EQUIPMENT		INCLUDED IN PROPOSAL
G2 PSA GENERAL		
One set of Proprietary and Patent-Protected, Multi-Port Rotary Control Valves		•
Nine Adsorbent Beds (Painted RAL 9016 "White") and first load of adsorbent media		•
Motor, Gearbox, Shaft, Shaft-Guard, and proximity switch for indicating valve rotation		•
Variable Speed Drive and control panel		•
Piping/tubing connections and flanges to Battery Limits		•
Interconnecting Process Tubing and Piping		•
One pressure relief valve on each PSA bed		•
Terminal Box JB1		•
Skid-mounted Frame		•
First Bed Pressure Sample Port with Pressure Gauge and Isolation Valve		•
FEED		
Explosion-proof Solenoid Isolation Valve		•
Pressure Gauge with 2 (1/2" NPT) Sample Ports (or for N2 Purge) and Isolation Valve		•
Coalescing Filter (0.01 micron)		•
Auto-float Liquid Drain Trap including 3-way T-Port Isolation Valve		•
Drain Isolation Valve		•
PRODUCT		
Explosion-proof Solenoid Isolation Valve		•
Particulate Filter (1 micron)		•
Pressure Gauge with 2 (1/2" NPT) Sample Port and Isolation Valve		•
Check Valve		•
Product Direct-Operated Back-Pressure Valve		•
PSA product purge line with back pressure regulator, pressure gauge and manual isolation valve		*
Light recycle port with manual isolation valve		**
EXHAUST		
Manual Isolation Valve		•
Pressure Gauge with 2 (1/2" NPT) Sample Port and Isolation Valve		•
Manual Low Point Drain Valve		•
DOCUMENTATION		
One electronic copy of the Installation, Operating, & Maintenance Manual (in English)		•

* Product purge line is included for high product purity applications (>99.5% vol)

** Light recycle line is included for biogas/high recovery systems and must be piped back to biogas compressor suction line at less than 1.5 Psig pressure.

PSA GAS PURIFICATION SYSTEM

1.2.2 Available Options (Not included):

AVAILABLE
OPTIONS

COLD WEATHER PACKAGE



The function of the cold weather package is for freeze protection and limiting condensate near or sub-zero ambient conditions. Feed and exhaust lines are heat traced with self-regulating heating cables that are insulated with a flexible elastomer and aluminum cladding. Minimum operating ambient temperature specification will be changed from the standard specification of +4°C (+39°F) to -25°C(-13°F). Heat trace cables are CSA, UL, & FM approved for Class 1, Division 2, Groups A, B, C, and D service.

An additional 2 weeks is required for delivery (to install the cold weather package.)

EXHAUST SURGE TANK OPTION



The function of the exhaust surge tank is to dampen pressure and caloric value fluctuations of the exhaust stream before entering a burner or similar pressure pulsation sensitive equipment. The exhaust line will be connected to a stand-alone, off-skid, surge tank to achieve $\pm 5\%$ variation in Wobbe index at Purchaser's operating conditions. The surge tank will be painted RAL 9016 Traffic White (or Apollo White PS 39/10210). Option includes a safety pressure relief valve (relief pressure setting at 14.5 PSIG).

PRODUCT FLOWMETER OPTION



The product flowmeter option provides product flow indication. The differential pressure, static pressure, and process temperature are fully compensated to measure mass flow "real-time" using a multi-variable mass flow transmitter. The complete assembly includes an integrated orifice manifold that further improves accuracy and repeatability for reliable product flow measurements. The flowmeter is FM (default selection) approved or CSA approved (if requested by Purchaser) for Class 1, Division 1, Groups B, C, and D service.

PRESSURE RELIEF VENT COLLECTION MANIFOLD

The function of the pressure relief vent collection manifold is to collect the vent gas that is discharged from the pressure safety valves. For indoor operations, the pressure relief vent header is recommended. A common tie-in point (150# flange) is provided for pipe connection to a safe location.

ENHANCED PRODUCT PURGE CONTROL



The function of the enhanced product purge control is to optimize performance by increasing recovery at turndown, product purge is automatically tuned to use the optimal amount of purge gas for regenerating the adsorbent beds. Enhancement is done using a mass flow controller via a PLC. This option replaces the standard needle valve and pressure reducing valve on the product purge line. The expected gain in recovery at turndown can be up to 3 percentage points of recovery (*actual gain to be confirmed with Xebec*). The mass flow controller is UL or FM approved for Class 1, Division 2, Group B service.

PSA GAS PURIFICATION SYSTEM

AVAILABLE
OPTIONS
(cont.)

ADVANCED CONTROL PACKAGE



The function of the Advanced Control Package is to allow for safe unattended, stand-alone, auto-control operation of the PSA that includes a PLC and a full-color touch-screen HMI for full PSA control (with trending/data logging functionality). PLC I/O is expandable to accommodate user's needs and optional communication connection available to the Purchaser's PLC. Control logic is pre-loaded and fully customizable. All electrical components are contained within an enclosed NEMA 4 (IP 67) cabinet. The cabinet is to be installed by Purchaser in a non-classified area within 30 m (100 ft) of the PSA where ambient temperatures are between 4-50°C.

FILTER BYPASS



The function of the filter by-pass is to allow replacement of the coalescing filter element without the need to shutdown the entire system. Two isolation ball valves for the filter and an isolation ball valve on the bypass line are provided. This option is recommended for applications with anticipated high levels of particulates and water droplets that would decrease the service life of the filter element.

PRODUCT BYPASS



The function of the product by-pass line is used to divert off-specification product from the product line to a designated discharge location. The product bypass line can also be used for initial plant startup or when the product isolation valve is closed. The product bypass line comes with a direct operated back-pressure valve, pressure gauge with 2 (1/2" NPT) sample ports, and an isolation ball valve.

ELECTRO-PNEUMATIC PRESSURE CONTROL VALVE



The function of the electro-pneumatic product control valve is to provide backpressure control when the product downstream pressure is significantly lower than the feed upstream pressure. This option replaces the standard manual-adjusted backpressure valve. The downstream operating pressure is to be specified by the Purchaser with this option for initial C_v setting of the valve. Instrument air (60 PSIG) is required. The electro-pneumatic control valve comes with a transducer that is FM (default selection) approved or CSA approved (if requested by Purchaser) for Class 1, Division 1, Groups B, C, and D service. This option also includes an explosion proof pressure transducer that is UL & cUL listed for Class 1, Division 1, Groups A, B, C, and D service.

PRODUCT SURGE TANK OPTION



The function of the product surge tank is to dampen pressure fluctuations of the product stream. The product line will be connected to stand-alone, off-skid pressure vessel. The size of the surge tank is given in Section 1.2. The surge tank will be painted RAL 9016 Traffic White (or Apollo White PS 39/10210). Option includes a safety pressure relief valve (relief pressure setting at relief pressure given in Section 1.2). The surge tank will be designed to ASME Section VIII, Division 1.

EXTERNAL PRODUCT PURGE



The function of the external product purge line is to provide an additional product gas purge step to assist in the further regeneration of the adsorbent beds. The added benefit of this feature is higher performance for selected applications. The external product purge line comes with a direct operated back-pressure valve, needle valve, pressure gauge with 2 (1/2" NPT) sample ports, and an isolation ball valve. This option is always included for high purity hydrogen applications.

1.3 MAINTENANCE & RECOMMENDED SPARE PARTS

Prior to delivery, the G2 PSA will be pressurized with helium or air to ensure containment, checked for valve rotation and synchronization, tested for mechanical operation, and tested for other electrical connections.

A summary of the recommended spare parts and maintenance intervals are provided below as a general guideline. A detailed inspection and preventive maintenance chart is provided in the Installation, Operation, and Maintenance manual.

MAINTENANCE	PERIODIC INSPECTION (10 minutes / inspection) Periodic inspection is recommended for preventative maintenance. Motor, gearbox, and valves are inspected for irregular noises, leaks, and obvious degradation of connections (e.g. – grounding, electrical, brackets). Coalescing filter is inspected and condensate is drained at various collection points. Inspection intervals will vary from every 2-3 days to weeks depending on the severity of service and environmental conditions.
	2.5 YEAR SERVICE (2 day / 2.5 years) 2.5 Year Service of the complete system is recommended for optimal performance tuning and overall system adjustment. Internal components of the rotary valve are inspected and parts are replaced as required. Instruments and equipment are tested and checked for accuracy. Filter element is replaced as required. All motor, gearbox, and mechanical couplings are lubricated and/or oil changed as required. Recommended 2.5 Year Service is to be done by Xebec approved personnel.
	5 YEAR SERVICE (2 day / 5 years) Various internal components of the rotary valve are replaced, cleaned, and corrected for alignment. The 5-year service is to be done by Xebec approved personnel only.
SPARE PARTS	ROTARY VALVE SPARE PART KIT (recommended - price available upon request) Valve Seals and Service Kit
	GENERAL SPARE PART KIT (recommended - price available upon request) Variable Speed Drive (VSD); Filter Elementsfor Xebec supplied filters; Two Regulator Service Kits (for back-pressure valves)

1.4 ENVIRONMENTAL ASPECTS

- The unit will not produce any gaseous or liquid effluent other than the Product and Exhaust gases (and moisture condensate from feed gas in some installations).
- Solid effluents will consist of used filter cartridges only. The Purchaser is to dispose effluents in satisfactory manner and in compliance with all applicable laws including environmental laws. Purchaser covenants and agrees with Xebec to indemnify Xebec against all liabilities, claims, demands, actions, causes of action, damages, losses, costs or expenses (including legal fees and disbursements) suffered or incurred by Xebec by reason of or arising out of Purchaser's breach of this covenant.

SECTION 2: FINANCIAL PROPOSAL

2.1 PRICE AND COST SCHEDULE

2.1.1 Xebec Scope of Supply & Price:

SCOPE OF SUPPLY & PRICE		PRICE (\$ USD)	NOTES & EXCEPTIONS
STANDARD FEATURES & SPECIFICATIONS (as defined in Section 1.2)			
PSA3H-G2-330A-480		\$93,000.00	
OPTIONS (not included in price and as defined in Section 1.2.2)			
Cold Weather Package		Avail. Upon request	
Exhaust Surge Tank		Avail. Upon request	
Product Flowmeter		Avail. Upon request	
Pressure Relief Vent Collection Header		Avail. Upon request	
Enhanced Product Purge Control		Avail. Upon request	
Advanced (Control Package)		Avail. Upon request	
Filter Bypass		Avail. Upon request	
Product Bypass		Avail. Upon request	
Electro-Pneumatic Product Control Valve		Avail. Upon request	
Product Surge Tank Option		Avail. Upon request	
External Product Purge Option		Avail. Upon request	
Installation, Operating, & Maintenance Manuals	One electronic copy in English	Included	
	Additional Electronic Copies	\$100 ea.	
	Printed Copies	\$200 ea.	
SPARE PARTS (as defined Section 1.3)			
Rotary Valve Spare Part Kit		Avail. Upon request	
General Spare Part Kit		Avail. Upon request	
FREIGHT & TRANSPORTION			
Delivery Time after receipt of first milestone payment pending production availability and based against 5 business days for client approval of prelim engineering documents.		20 to 24 weeks	
Xebec Standard Packaging		Included	
Shipping Terms		Exworks Xebec Blainville factory, Quebec, Canada	
Insurance		By Purchaser	

Taxes and duties are extra. Minimum order is 1 unit(s).
This proposal is valid for a period of three (3) months.

The unit will be ready to ship at the proposed delivery time above, provided the purchaser has approved all Xebec supplied drawings and equipment by week 4 for custom orders. Customer approval of Xebec drawings and documents are not required for standard orders.

Although every effort is made at the quotation stage to include the most acceptable materials and latest designs for our products, Xebec reserves the right to alter, modify, or change these systems as necessary to conform to its latest designs at the time of order. Any changes will be conveyed to the Purchaser. Should the Purchaser require any modification, alteration, or change, supplier reserves the right to adjust the proposal prices or deliveries accordingly.

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2.1.2 Product Support:

PRODUCT SUPPORT	COMMISSIONING BY XEBEC (to be quoted separately) In accordance to our latest service rate plus mobilization charges including transport, accommodation, based on a 7.5 hour day, 5 day week, Monday to Friday. Additional hours and weekends/holidays will be charged at premium rate.
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2.2 PURCHASER SCOPE OF SUPPLY

PURCHASER SCOPE OF SUPPLY	EQUIPMENT • Level concrete pad for foundation and anchor bolting. • Piping and fittings for connection to PSA. • PSA exhaust collection system to provide PSA exhaust pressure of Section 1.1. • Overpressure protection for feed conditions above design conditions in Section 1.1. • Control system interface/signals as required by Xebec PSA control logic. • Provision for gradual feed pressurization used on initial start-up. • Loading of adsorbent media in the PSA vessels. • Utilities Hook up, nitrogen purge (for maintenance), and instrument air (for electro-pneumatic product control valve option) by Purchaser. • Installation of the panel CP1 in a non-hazardous (non classified) area, with A/C or good air circulation, electrical wiring of CP1 to his controller and power source, electrical wiring of CP1 to the PSA motor. CP1 contains mainly the VFD, power supply and pre-wired connection terminals on which the purchaser should connect the remote signal of his controller (PLC, DCS or else). The panel may contain other items depending on the selected options. The purchaser is responsible to provide the power source and protection to the panel. The panel should be located at no more than 50m from the PSA motor. • Electrical wiring of Junction boxes (JB1 and JB2) to purchaser's controller (PLC or DCS), JB1 and JB2 are installed on the PSA skid and pre-wired (by Xebec) to the devices of the PSA (SV, proximity, etc) • Site preparation, off-loading and installation. • Any PSA equipment integration, site evaluation, process hazard studies, risk assessment studies (e.g. – FMEA, HAZOP)
	TESTING • Pressure testing of connected system. • Correct rotation of motor. • Correct connection of electrical components. • Control System and I/O checkouts • Correct programming of control logics (supplied by Xebec) including fail-safe protection trip.
	ADMINISTRATIVE • Permits, licenses and approvals. • Fire protection systems. • Any item not detailed in this document. • Consumables.

2.3 CREDIT TERMS:

Credit is approved for Purchaser to the Terms of Payment (Section 2.4), subject to ongoing review.

2.4 TERMS OF PAYMENT

2.4.1 Payment Schedule:

	<u>Payment Due:</u>
40% (forty) of equipment price:	Upon receipt of a purchase order
40% (forty) of equipment price:	Upon submission of preliminary GA, P&ID and electrical schematics
20% (twenty) of equipment price:	Upon notice of readiness to ship
Commissioning price:	to be quoted separately

Note: In order for the performance warranty to be valid, the equipment must be started by a Xebec authorized service technician.

2.4.2 Payment Details:

Late payment charges are 2% per month. All payments are to be considered non-refundable deposits. In the event that payment is not received within 30 days of the above terms, Xebec has the option to terminate this Agreement.

2.5 CONDITIONS OF CONTRACT:

2.5.1 Conditions

Any purchase order issued by Purchaser in respect of this Proposal will be subject to acceptance by Xebec and the terms and conditions of this Proposal, and, in the event of any inconsistency between this Proposal and the purchase order, this Proposal will govern unless Xebec expressly consents in writing to a term of the purchase order overriding this Proposal. If any purchase order introduces a term or condition not covered by this Proposal, then such term or condition shall not bind Xebec unless Xebec expressly consents in writing to such term or condition.

2.5.2 Standard Conditions:

Xebec Adsorption "General Terms and Conditions of Sale" shall apply where not specified in this proposal.

SECTION 3: GENERAL TERMS AND CONDITIONS OF SALE

Shall constitute a part of Xebec's standard sale agreement

3.0 Definitions:

3.0.1 "Goods":

All the agreed upon items and associated ancillary equipment that constitutes due fulfillment of this project.

3.0.2 "Supply":

To organize, design, manufacture, purchase, expedite, assemble, and test all the "Goods" F.O.B., at Xebec's premises, in either fully assembled, partially assembled or fully dismantled form.

3.0.3 "Transport":

To organize and provide suitable transport for the delivery of all the "Goods" from Xebec's premises to the Purchaser's site.

3.0.4 "Inspection":

To witness, by a Purchaser's authorized representative, suitable "factory testing" at the Xebec's premises of all "Goods".

3.0.5 "Installation":

To set out, place in position, line up, and connect up all "Goods" to other installations or equipment, and to plant services provided, using the Purchaser's installation team.

3.0.6 "Commissioning":

To provide a Xebec certified technician, or other such suitably qualified person, for the tuning of mechanical and electrical components and putting the "Goods" in operational mode to allow commercial and/or testing use.

3.1 Prices:

Unless otherwise specified by Xebec, Xebec's price for the Goods shall remain in effect for two months after the date of Xebec's quotation or acceptance of the order for the Goods, whichever is delivered first, provided an unconditional, complete authorization for the immediate manufacture and shipment of Goods is received and accepted by Xebec within such time period.

3.2 Delivery and Documentation:

All shipping dates are approximate and are based upon Xebec's prompt receipt of all necessary information from the Purchaser to properly process the order. Unless otherwise specified by Xebec, the Purchaser shall pay transportation and handling expenses. Delivery to carrier shall constitute passage of risk of loss to the Purchaser irrespective of arrangements for transportation or insurance charges.

3.3 Shortage:

The Purchaser must inspect Goods promptly upon receipt and must submit any claim for shortage within thirty days after receipt or any such claim shall be waived.

3.4 Cancellation by Purchaser:

If Purchaser cancels this agreement or refuses delivery of the Goods, the Purchaser shall be liable for and shall pay to Xebec all expenses incurred or committed to by Xebec prior to cancellation and Xebec's profit to the same extent as if the Purchaser had accepted delivery of the finished Goods.

3.5 Force Majeure:

Xebec shall not be liable for delays in performance or for non-performance due to acts of God, war, riot, fire, labour troubles, unavailability of materials, components, or labour, explosion, accident, storm, flood, earthquake, compliance with governmental requests, laws, regulations, orders or actions, or unforeseen circumstances or causes beyond Xebec's reasonable control.

3.6 Operational and Process Performance Testing

Testing will be performed in the following steps:

PSA GAS PURIFICATION SYSTEM

3.6.1 Mechanical Inspection and Operational Test

- (a) A Mechanical Checkout and Operational Test will be performed on the Goods at Xebec's factory prior to shipment.
- (b) No fluids processing will be performed during this step.
- (c) Purchaser representatives will be invited to see this test. Their travel and living expenses will be at the Purchaser's charge.
- (d) The test will be deemed successful if Xebec's Quality Department ensures all components of the Goods are present, properly installed and operational according to the Scope of Supply in Section 1.2.

3.6.2 Process Performance Test

- (a) A Process Performance Test will be performed at Purchaser's site under Xebec representative's supervision following Goods installation, systems checkout and start-up to prove achievement of the guaranteed performance criteria.
- (b) Purchaser will give fifteen (15) days of advance notice to the Xebec representative of the Process Performance Test starting day. Such test starting date must be within ninety (90) days after delivery of the Goods. Xebec will send qualified personnel to assist with the test. If the Purchaser fails to make the system available for Process Performance Test with ninety (90) days, the system will be deemed successfully complete and accepted by the Purchaser. The Purchaser will take over the Scope of Supply and the Purchaser will release Xebec from Process Guarantee. If no process guarantee is provided with the final quotation, then the process acceptance is deemed complete upon start-up.
- (c) Purchaser must provide all specified utilities and site requirements. Fluids, manpower, energy and analysis will be at Purchaser's charge. Analysis of the gas purity will be performed at a mutually acceptable test facility. Any flow measurement, purity analysis or other equipment required to accurately determine process performance will be at Purchaser's charge. The performance test will be subject to the accuracy limitations of those instruments. The Purchaser will provide calibration certificates and accuracy specifications for the instruments required, to the satisfaction of Xebec, prior to the Performance Test.
- (d) A twenty-four (24) hour or shorter mutually agreed test period of continuous operation test will be performed and all operating characteristics will be recorded and presented in report form to Purchaser. This test will be deemed successful if the process guarantee conditions specified in section 1 are met.
- (e) If conditions exist at the site beyond parties' control that prevent the successful performance of the test, parties will mutually agree to a different test schedule.
- (f) Upon completion of a successful Process Performance Test, both parties will sign a certificate of successful performance of the Process Performance Test; and the Purchaser will take over the Scope of Supply and the Purchaser will release Xebec from Process Guarantee.
- (g) In case the Goods fail to reach the guaranteed performance by reasons for which Xebec is responsible, Xebec's sole responsibility shall be, at its option either, (i) commence actions to carry out, on a best effort basis, at his own costs and within 1 (one) month after the Process Performance Test or such other time as the parties shall agree to, all remedial works that are necessary to achieve the guaranteed performance in a further Process Performance Test or (ii) refund the purchase price paid by Purchaser.

3.7 Warranty:

Xebec warrants the Goods against defects in workmanship or materials under normal use for twelve months from date of start-up or eighteen months from date of shipment, whichever is earlier. All replacements or repairs necessitated by inadequate preventive maintenance, or by normal wear and usage, or by fault of the Purchaser, or by unsuitable power sources or by attack or deterioration under unsuitable environmental conditions, or by abuse, accident, alteration, misuse, improper installation, modification, repair, storage or handling, or any other cause not the fault of Xebec ("Non-warranty Repairs") are not covered by this Limited Warranty, and shall be at Purchaser's expense. All costs of dismantling, reinstallation and freight and the time and expenses of Xebec's personnel for site travel and non-warranty repairs shall be borne by Purchaser unless accepted in writing by Xebec. Xebec's sole responsibility shall be, at its option, to repair or replace and install free of charge any parts or equipment found to be defective, up to a maximum of 50% of the purchase price, or refund the full purchase price paid by the Purchaser. Goods repaired and parts replaced during the warranty period shall be in warranty for the remainder of the original warranty period or ninety (90) days, whichever is longer. THE FOREGOING WARRANTIES AND REMEDIES ARE EXCLUSIVE AND EXPRESSLY IN LIEU OF ALL OTHER WARRANTIES EXPRESSED OR IMPLIED; AND XEBEC MAKES NO REPRESENTATION OR WARRANTY, EXPRESS OR IMPLIED, AS TO THE FITNESS OF A PARTICULAR PURPOSE OR MERCHANTABILITY OF THE GOODS.

3.7.1 Equipment Access

After providing reasonable notice to the Purchaser, Xebec personnel must have access to the purchaser site to collect performance data and samples during the warranty period.

3.8 Limitation of Remedy and Liability:

XEBEC SHALL NOT BE LIABLE FOR DAMAGES CAUSED BY DELAY OR PERFORMANCE. THE SOLE AND EXCLUSIVE REMEDY FOR BREACH OF WARRANTY HEREUNDER SHALL BE LIMITED TO REPAIR, CORRECTION OR REPLACEMENT UNDER THE LIMITED WARRANTY CLAUSE IN SECTION 3.7. IN NO EVENT, REGARDLESS OF THE FORM OF THE CLAIM OR CAUSE OF ACTION (WHETHER BASED IN CONTRACT, INFRINGEMENT, NEGLIGENCE, STRICT LIABILITY, OTHER TORT OR OTHERWISE), SHALL XEBEC'S LIABILITY TO PURCHASER EXTEND TO INCLUDE INCIDENTAL, CONSEQUENTIAL OR PUNITIVE DAMAGES. THE TERM "CONSEQUENTIAL DAMAGES" SHALL INCLUDE, BUT NOT BE LIMITED TO, LOSS OF ANTICIPATED PROFITS, LOSS OF USE, LOSS OF REVENUE, AND COST OF CAPITAL.

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3.9 Installation:

The Purchaser shall be responsible for receiving, storing, installing, starting up, and maintaining all Goods. Xebec shall provide a quotation for services to assist the Purchaser in these functions if requested. In order for the warranty to be valid, commissioning and start-up are to be performed by a Xebec certified technician.

3.10 Product Support:

Xebec's policy is that functionally interchangeable replacement parts will be available during the time a standard product is offered for sale. In addition, either repair capability, or functionally equivalent parts will be available for at least five years from the last date of product availability. No guarantee of parts availability, repair capability, or functionally equivalent goods is offered for goods manufactured by others and supplied with Xebec's Goods.

3.11 Taxes and Duties:

Xebec's prices do not include amounts for any duties or sales, use, excise, or similar taxes. The Purchaser shall pay, in addition to Xebec's prices, the amount of any present or future such tax and duty applicable to the sale or use of the Goods or, in lieu of such payment, provide Xebec with a tax exemption certificate and/or duty clearance acceptable to the appropriate authorities. The Purchaser covenants and agrees with Xebec to indemnify Xebec against all liabilities, claims, demands, actions, causes of action, damages, losses, costs or expenses (including legal fees and disbursements) suffered or incurred by Xebec by reason of or arising out of Purchaser's breach of this covenant.

3.12 Insurance:

The Purchaser shall provide and maintain insurance on the Goods in an amount not less than their full insurable value, with loss payable to Xebec and the Purchaser, as their interests may appear, from the date that risk of loss passes to the Purchaser until such time as title passes to the Purchaser and the Purchaser pays the full purchase price to Xebec.

3.13 Labeling:

Each of the Goods utilized by the Purchaser shall have a label affixed thereto identifying it as manufactured or developed by Xebec and containing such proprietary, copyright, patent, trademark, design right, trade secret and all other proprietary rights, legends and alpha-numeric codes as Xebec considers appropriate. The Purchaser shall ensure that such label remains affixed to such Goods, shall replace such label if it is destroyed, removed or becomes unreadable and in no event shall the Purchaser claim that the Goods or any component thereof has been developed by the Purchaser.

3.14 Milestone Payments:

Unless otherwise provided in Xebec's written quotation, the Purchaser shall make periodic milestone payments. Invoices shall be issued by Xebec and paid by the Purchaser based upon the following Milestones: Milestone 1: 40% of price upon receipt of a purchase order. Milestone 2: 40% of price upon submission of preliminary GA, P&ID and electrical schematics. Milestone 3: 15% of price upon notice of readiness to ship. Milestone 4: After commissioning or 60 days after shipment, whichever occurs first. All payments are non-refundable and due within 15 days of the invoice date. An interest charge of 2% per month will be assessed on late payments.

3.15 Purchaser Supplied Data:

To the extent that Xebec has relied upon any specifications, information, representation of operating conditions or other data supplied in writing by the Purchaser to Xebec in the selection or design of the Goods and the preparation of Xebec's quotation, and in the event that actual operating conditions or other conditions differ from those represented by the Purchaser and relied upon by Xebec, any warranties or other provisions contained herein which are affected by such conditions shall be null and void, unless otherwise mutually agreed upon in writing.

3.16 Storage:

The Purchaser agrees to pay Xebec a storage charge of 1% of the total purchase price per month, or prorated portion thereof for a part of a month, for Goods not authorized for shipment by the Purchaser within fourteen (14) days following either the specified delivery date or the date the Purchaser is notified that same are ready for shipment, whichever is later.

3.17 Title

(a) Purchaser agrees to grant to Xebec such security interests in the Goods and to execute such other documents as may be desirable for Xebec to obtain a perfected first ranking charge over the Goods as security for the payment of all monies owing to Xebec by the Purchaser and performance of all obligations of the Purchaser under this agreement.

(b) It will be a default if the Purchaser fails to make any payment due under this Agreement or any other agreement between the Purchaser and Xebec when due.

(c) Until Xebec has been paid in full for the Goods in accordance with this Agreement, title to the Goods shall remain exclusively in Xebec.

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(d) Purchaser acknowledges receipt of a copy of this Agreement and waives all rights to receive from Xebec a copy of any financing statement, financing change statement, or verification statement filed at any time at the Personal Property Registry or other similar registry in respect of this Agreement.

(e) In the event of default of this Agreement or in the payment of any sums due to Xebec as aforesaid, Xebec may exercise any and all remedies afforded to secured parties by Part 5 of the *Personal Property Security Act* or other similar statutes, including seizure of the Goods.

3.18 Proprietary Rights

(a) Xebec shall defend at its own expense any legal proceeding brought against the Purchaser and pay those damages finally awarded against the Purchaser in such proceeding, to the extent that it is based on a claim that the Goods in the form delivered to the Purchaser constitutes an infringement of any valid patent trademark, copyright, trade secret or other third-party proprietary right held under the law of the country in which the Site is located.

The Purchaser shall notify Xebec promptly in writing of any claim of infringement.

(b) Xebec shall have no liability or obligation to the Purchaser hereunder with respect to any patent, trademark, copyright, trade secret or other third-party proprietary right infringement or claims thereof based upon (i) unauthorized modifications of the Goods, (ii) any claim of patent, trademark, copyright, trade secret or other third-party proprietary right infringement in which the Purchaser or any affiliate or purchaser of Purchaser has an interest or license.

(c) In the event that any of the Goods is held to constitute infringement or the use thereof is enjoined, Xebec may at its option and at its own expense, as the exclusive remedy to the Purchaser therefor, either, (i) procure for the Purchaser the right to continue using such Goods, or (ii) replace such Goods with non-infringing Goods of equivalent quality and performance or, (iii) modify such Goods so that it is non-infringing, or (iv) accept return of such Goods.

(d) The foregoing states the entire liability of Xebec with respect to infringement of patents, trademarks, copyrights, trade secrets, or other third-party proprietary rights by any Goods delivered under this Agreement.

(e) The sale acceptance of the any Goods to the Purchaser by Xebec shall not confer on the Purchaser a license under any patent, trademark, copyright, trade secret or other proprietary rights of Xebec, and shall not confer on the Purchaser a license to reproduce, redesign, modify, create derivative works from, reverse engineer, disassemble or reverse assemble the Goods.

3.19 General Provisions:

(a) These terms and conditions, and the documents to which they are attached, constitute the entire agreement between the parties regarding the sale and purchase of the Goods and/or services to be performed and no other promises or agreements shall be of any force or effect unless mutually agreed upon in writing.

(b) In the case of contradictory terms or conditions of sale between this document and the attached documents which constitute the entire agreement, the terms or conditions of the attached documents shall take priority.

(c) These terms shall be binding upon and shall accrue to the benefit of the parties hereto and their respective assigns and successors in interest; provided, however, neither party may assign this agreement, or any of its interests herein, without the prior written consent of the other party.

(d) No action, regardless of form, may be brought by either party more than two (2) years after the cause of the action has accrued.

(e) The laws of the Province of Québec, Canada shall govern the validity, interpretation, and performance of this agreement. Canadian Courts shall adjudicate all disputes and all judgements shall be binding and final.

(f) Purchaser represents and warrants that Goods supplied hereunder are not to be used in nuclear applications, including, without limitation, any nuclear power generation facility.